

Linear model : interpreting the coefficients

Here is the English translation of your text, with LaTeX preserved and the language polished for clarity and precision :

Be careful when interpreting the coefficients, especially in the presence of collinearity among explanatory variables.

Suppose we have a set of explanatory variables, represented by the columns $X^1, \dots, X^p$ of the design matrix. We divide this set into two groups, say $\{X^1, \dots, X^q\}$ and $\{X^{q+1}, \dots, X^p\}$. Let $\hat{\beta}_1^{\text{complet}}$ denote the vector of coefficients associated with the first q variables in the **full model**, where all p explanatory variables are included. Let $\hat{\beta}_1^{\text{seules}}$ denote the vector of coefficients for the same q explanatory variables, but in the model where they are the only explanatory variables.

Let us now introduce some notation. Denote $\mathbf{X}_1$ as the matrix corresponding to the first q explanatory variables (i.e., the matrix $\mathbf{X}$ with its last $p - q$ columns removed), and $\mathbf{X}_2$ as the matrix corresponding to the remaining $p - q$ variables. We are then comparing the following two models :

$$\mathbf{y} = \mathbf{X}_1 \beta_1^{\text{complet}} + \mathbf{X}_2 \beta_2^{\text{complet}} + \epsilon$$

$$\mathbf{y} = \mathbf{X}_1 \beta_1^{\text{seules}} + \epsilon$$

We also define $M_2 = I - P_2$, where P_2 is the orthogonal projection onto the space spanned by the last $p - q$ explanatory variables. Hence, M_2 is the projection onto the orthogonal complement of that space. More generally, for any variable z , the residual vector of its regression onto the last $p - q$ explanatory variables is given by $M_2 z$.

Now, if we regress y only on the first q variables, ignoring the rest, we get :

$$\hat{\beta}_1^{\text{seules}} = (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T \mathbf{y}.$$

However, this vector $\hat{\beta}_1^{\text{seules}}$ does not coincide with the vector $\hat{\beta}_1^{\text{complet}}$, which contains the coefficients of those same q variables when the remaining $p - q$ variables are included in the model. And most importantly, the interpretation of the coefficients differs between the two cases.

Frisch–Waugh Theorem

1. The vector $\hat{\beta}_1^{\text{complet}}$ can be obtained by regressing y on the residuals $M_2 x^j$ for $j = 1, \dots, q$:

$$\hat{\beta}_1^{\text{complet}} = [(M_2 \mathbf{X}_1)^T M_2 \mathbf{X}_1]^{-1} (M_2 \mathbf{X}_1)^T \mathbf{y}.$$

2. Alternatively,

$$\hat{\beta}_1^{\text{complet}} = (\mathbf{X}_1^T \mathbf{X}_1)^{-1} \mathbf{X}_1^T (\mathbf{y} - \mathbf{X}_2 \hat{\beta}_2^{\text{complet}}).$$

In other words, $\hat{\beta}_1^{\text{complet}}$ can be obtained by regressing the part of $\mathbf{y}$ not explained by the last $p - q$ explanatory variables (i.e., the residual $\mathbf{y} - \mathbf{X}_2 \hat{\beta}_2^{\text{complet}}$) on the first q variables.

Comments on item 2

- If the columns are orthogonal, i.e., $\mathbf{X}_1^T \mathbf{X}_2 = 0$, then $\hat{\beta}_1^{\text{complet}} = \hat{\beta}_1^{\text{seules}}$. When the columns are centered, this corresponds to the case where the columns are uncorrelated.
- In the general (non-orthogonal) case, the coefficient of an explanatory variable x_j will depend on the model in which it is computed : **it will typically differ depending on whether certain other explanatory variables are included in the model, especially those that are correlated with x_j .**

Comments on item 1 Let us consider the particular case where $q = 1$, i.e., we isolate the first explanatory variable. Then item 1 states that, to compute the coefficient of this variable in the full model, we can project it onto the orthogonal complement of the space spanned by the other variables, and then regress y on that projection :

$$Z^1 = X^1 - P_{\mathbf{X}_2} X^1,$$

$$\hat{\beta}_1^{complet} = \left\langle \frac{Z^1}{\|Z^1\|_2^2}, \mathbf{y} \right\rangle.$$

(The same holds for any explanatory variable.)

Thus, in a multiple linear regression, the j th coefficient $\hat{\beta}_j$ represents the additional contribution of x_j to y , after adjusting x_j for the other explanatory variables.

Aside : Using the formula above, we immediately get :

$$Var(\hat{\beta}_j) = \frac{\sigma^2}{\|Z^j\|_2^2}.$$

This shows that if $\|Z^j\|_2$ is very small, the variance of the coefficient $\hat{\beta}_j$ becomes very large. This occurs when the variable X^j is highly correlated with the other explanatory variables.

Illustration : see the notebook "Frisch-Waugh".